

8257 Direct Memory Access Controller (DMAC) — MPCA

Module 4

Definition: The **Intel 8257** is a 4-channel programmable **Direct Memory Access Controller (DMAC)** that enables high-speed data transfer directly between I/O peripherals and system RAM without CPU intervention, bypassing the slow CPU fetch-execute cycle.

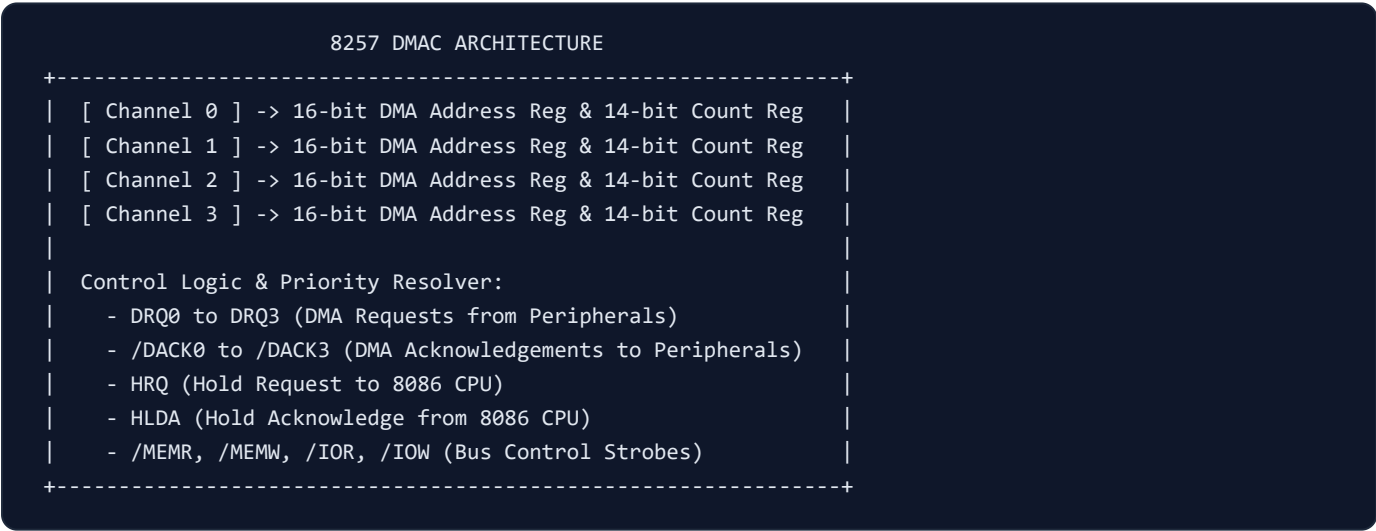
1. Detailed Technical Explanation

1. The DMA Concept & Speed Advantage

In standard programmed I/O, every byte transferred from a disk/network card to RAM requires multiple CPU instructions (`IN AL, DX` followed by `MOV [SI], AL`). DMA bypasses the CPU:

PROGRAMMED I/O:	Peripheral ---> CPU (AL Reg) ---> RAM	(Slow: ~20-50 Clock cycles/byte)
DMA TRANSFER:	Peripheral =====> RAM	(Fast: 2-4 Clock cycles/byte)

2. 8257 Internal Block Diagram & Channels



3. DMA Transfer Sequence (Step-by-Step)

- Peripheral asserts **DMA Request (DRQ)** to 8257.
- 8257 asserts **Hold Request (HRQ)** to the 8086 CPU.
- 8086 finishes its current bus cycle, floats its Address, Data, and Control buses into high-impedance state, and asserts **Hold Acknowledge (HLDA)** to 8257.
- 8257 takes master control of system buses, sends **DACK** to peripheral.

5. 8257 places memory address on address bus and issues simultaneous I/O Read and Memory Write (or Memory Read and I/O Write) to transfer data in a **single bus cycle**.
6. After count register reaches zero (**Terminal Count TC**), 8257 lowers **HRQ**, returning bus mastership to 8086 CPU.

4. DMA Transfer Modes

1. **Single Transfer Mode:** 8257 transfers one byte per request, releasing the bus back to CPU between transfers.
2. **Block Transfer Mode (Burst Mode):** 8257 transfers the entire block of data continuously until Terminal Count is reached.
3. **Demand Transfer Mode:** 8257 continues transferring data as long as peripheral holds **DRQ** active.
4. **Cascade Mode:** Allows multiple 8257 master-slave chips to be interconnected to expand channels.

5. Must-Write Points for Exams

- 8257 has **4 independent DMA channels** (Channel 0 to Channel 3).
- Each channel contains a **16-bit Address Register** and a **14-bit Terminal Count Register**.
- DMA transfer achieves maximum memory-to-I/O bandwidth with zero CPU software overhead.

6. Quick Recall Flow

DRQ -> HRQ -> CPU floats bus & asserts HLDA -> DACK -> Single Cycle MEM/IO Read/Write -> Terminal Count (TC)